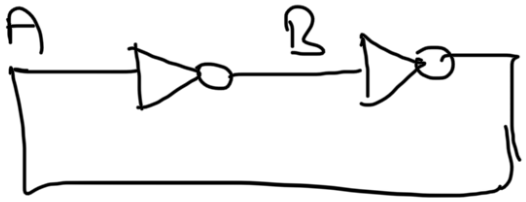
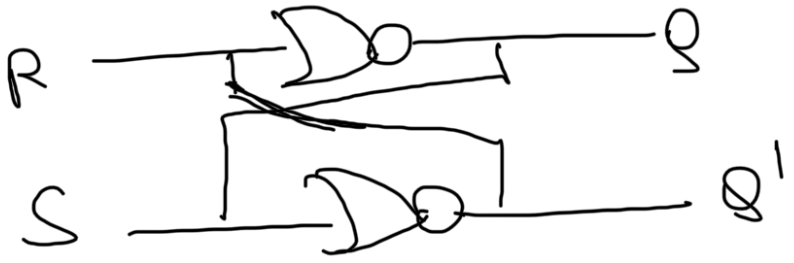
Basic Storage element.

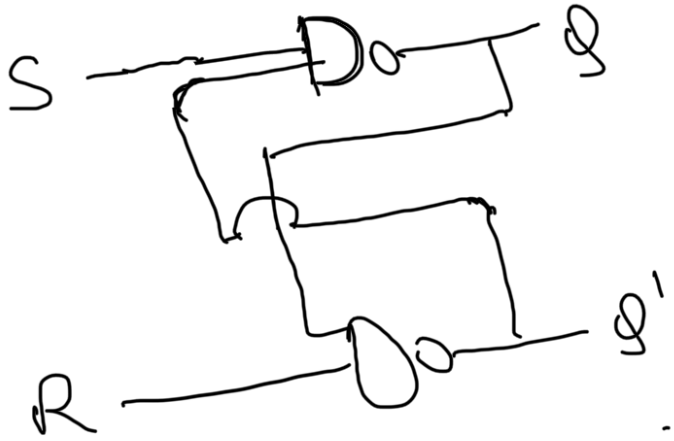


Latch supports set & reset



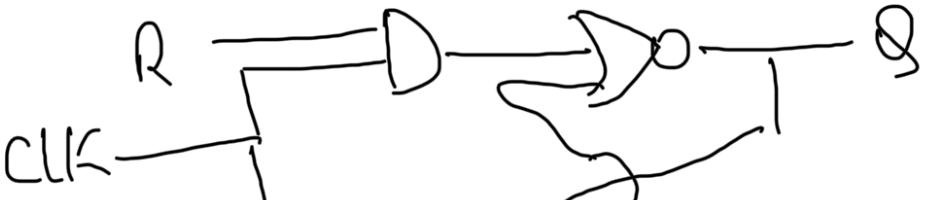
S	R	Q	Q'
1	0	1	0
0	0	0	0
0	1	0	1
0	0	0	0
1	1	forbidden	

memory



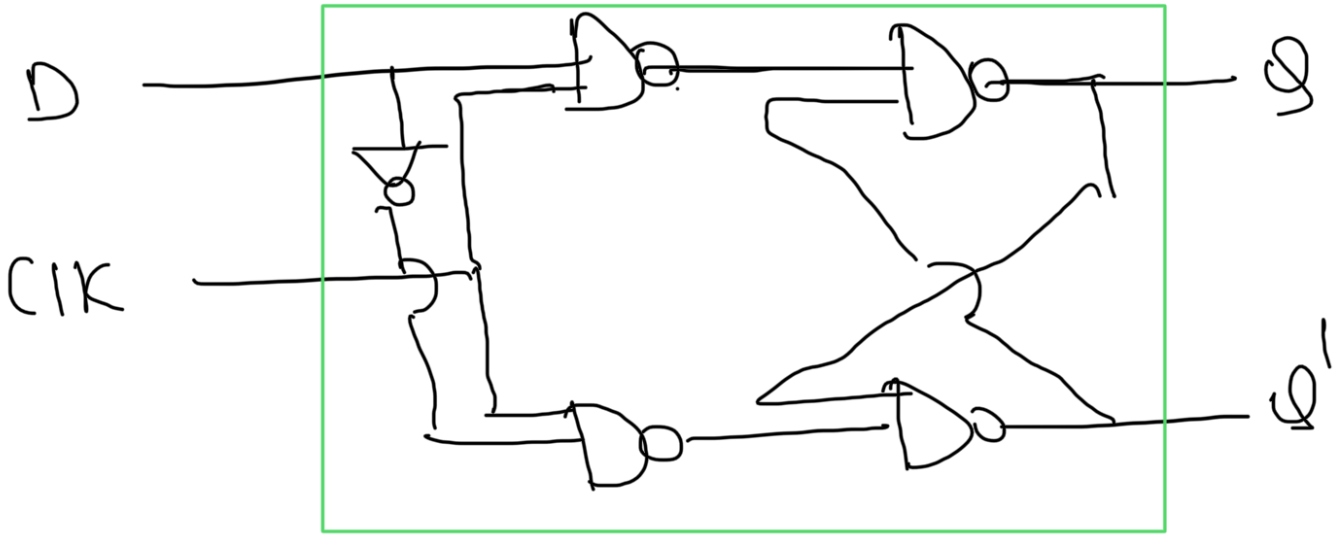
S	R	Q	Q'
1	0	0	1
1	1	0	1
0	1	1	0
1	1	1	0
0	0	forbid	

memory

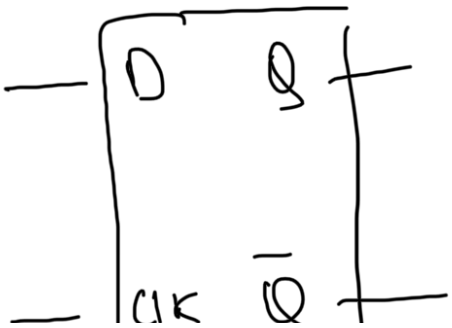




CLK	S	R	Q(t+1)
0	x	x	memory
1	0	0	memory
↓	0	1	0
↓	1	0	1
↓	↑	↑	not allowed



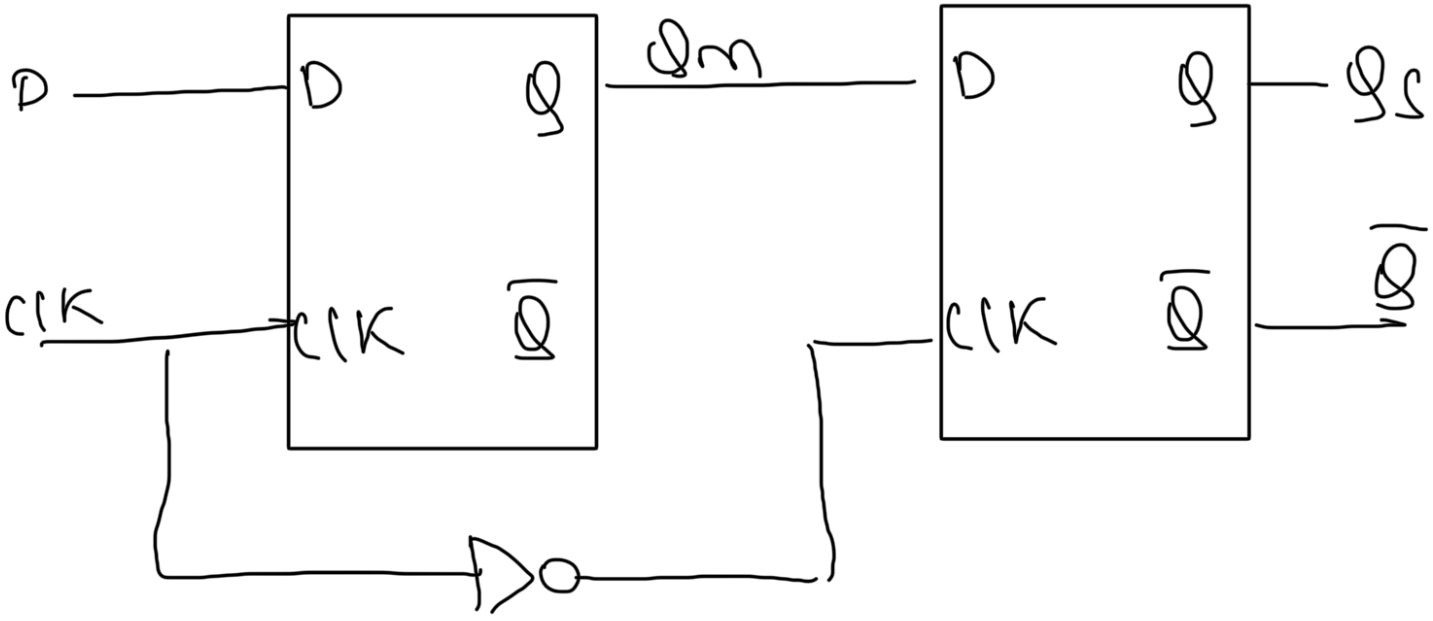
CLK	D	Q(t+1)
0	x	memory
↓	0	0



→ level sensitive

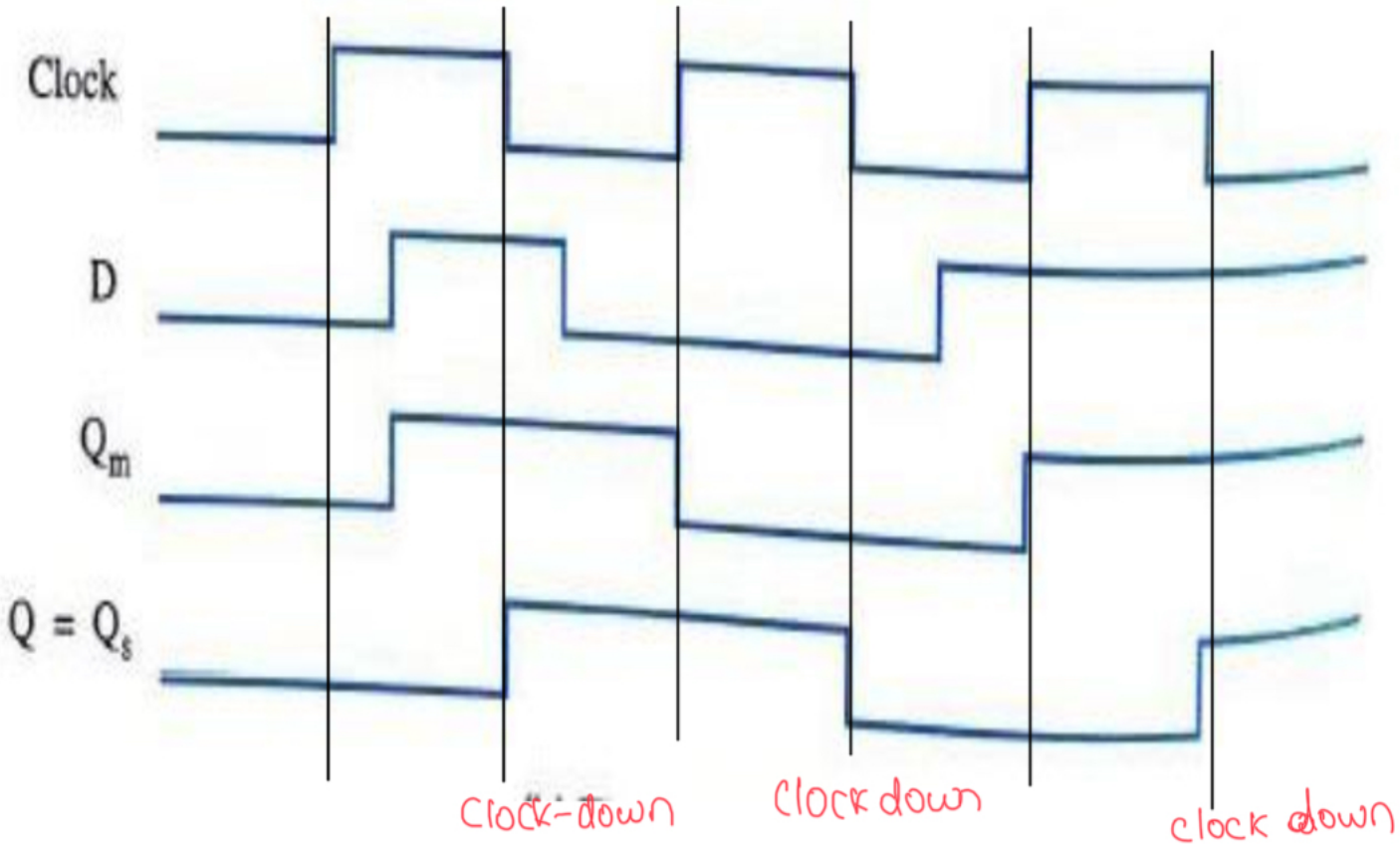
we want storage elements that change their States no more than once in a clock cycle.

master-slave D flip flop

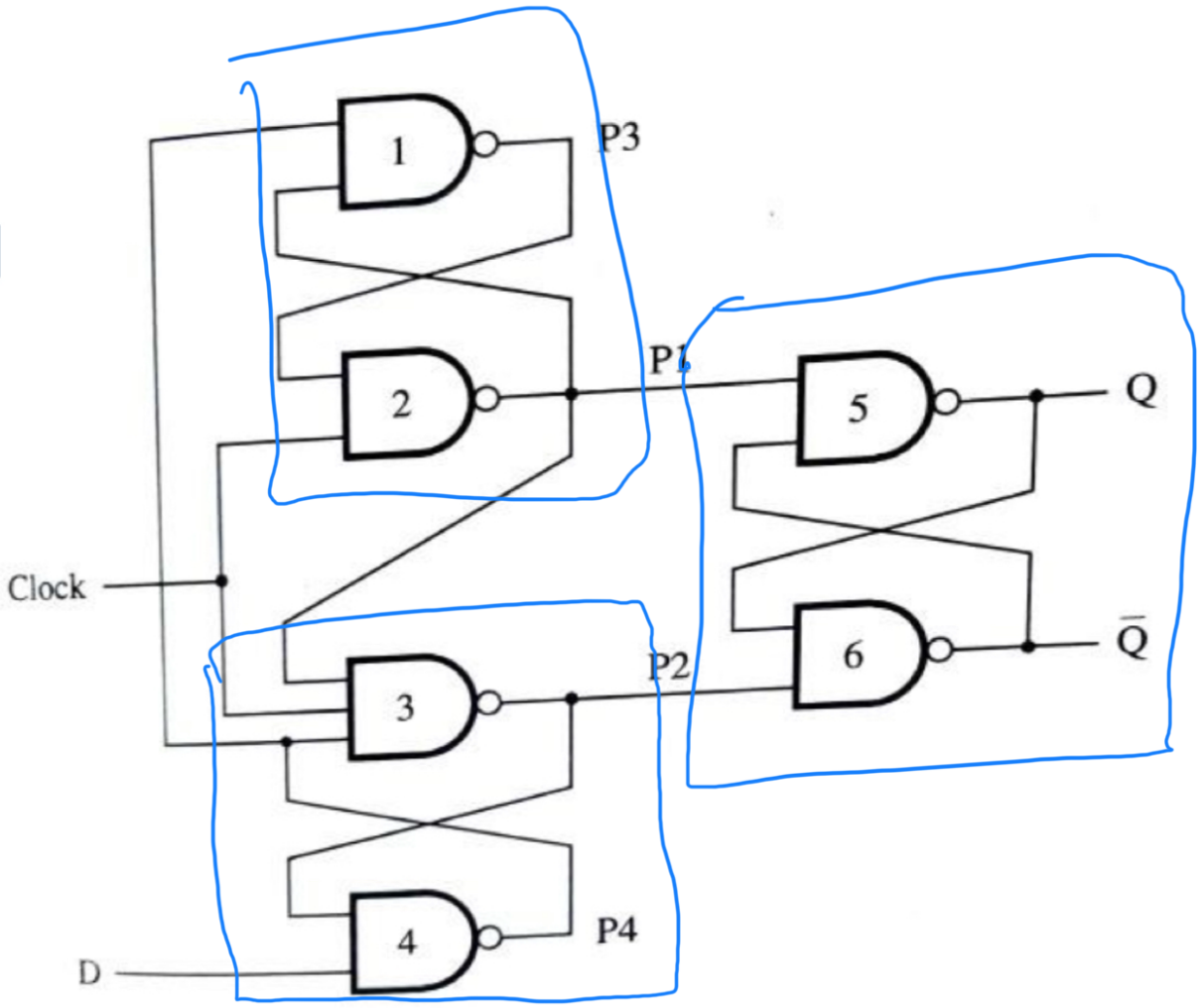


(negative edge triggered)

(Negative Edge Triggered)



Positive edge triggered D.ft



CLOCK = 0 → 2,3 → high ⇒ $P_1, P_2 = 1$

↓

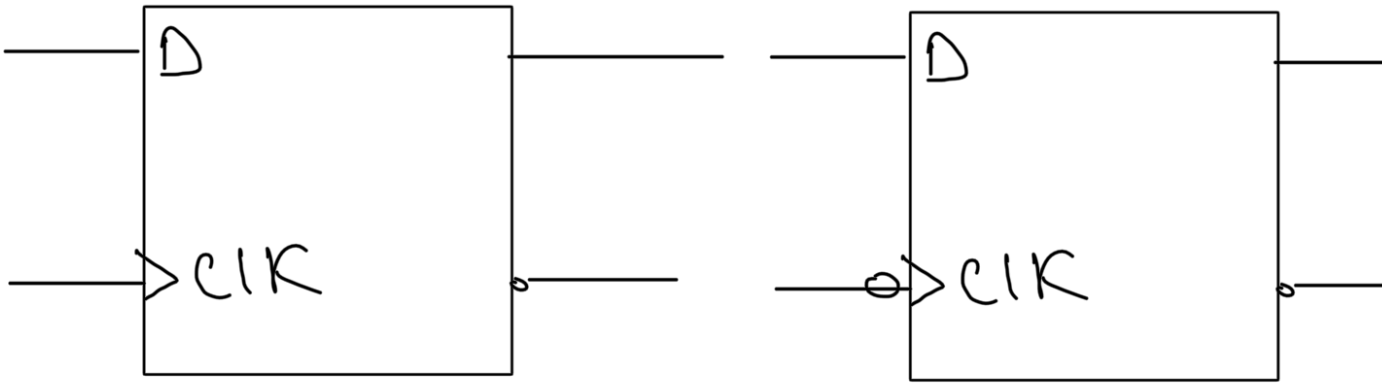
Q → maintained

$P_3 = D$

$P_4 = \bar{D}$

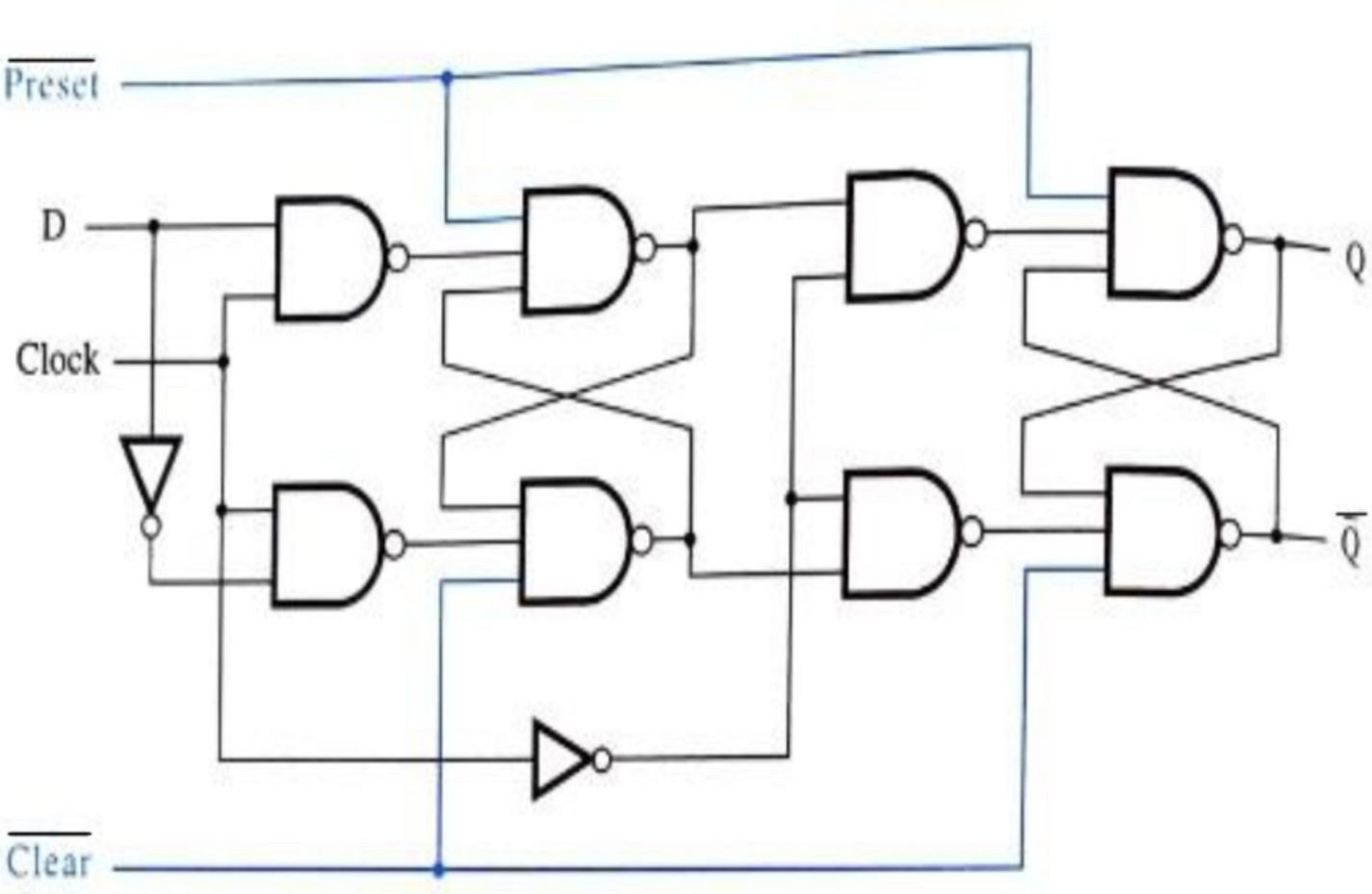
CLOCK : 0 → 1 ⇒ $P_1 = \bar{D}$ $P_2 = D$

⇒ $Q = D$, $\bar{Q} = \bar{D}$



+ve edge

-ve edge

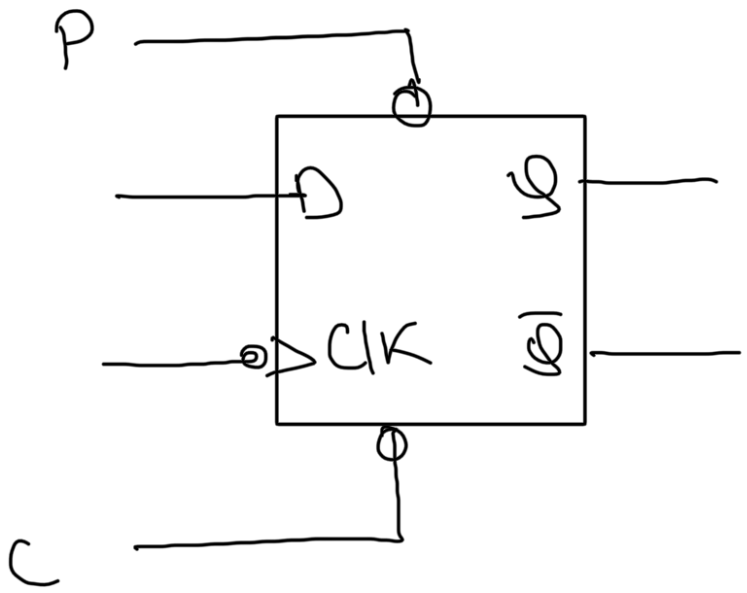


preset = 0 ⇒ Q = 1

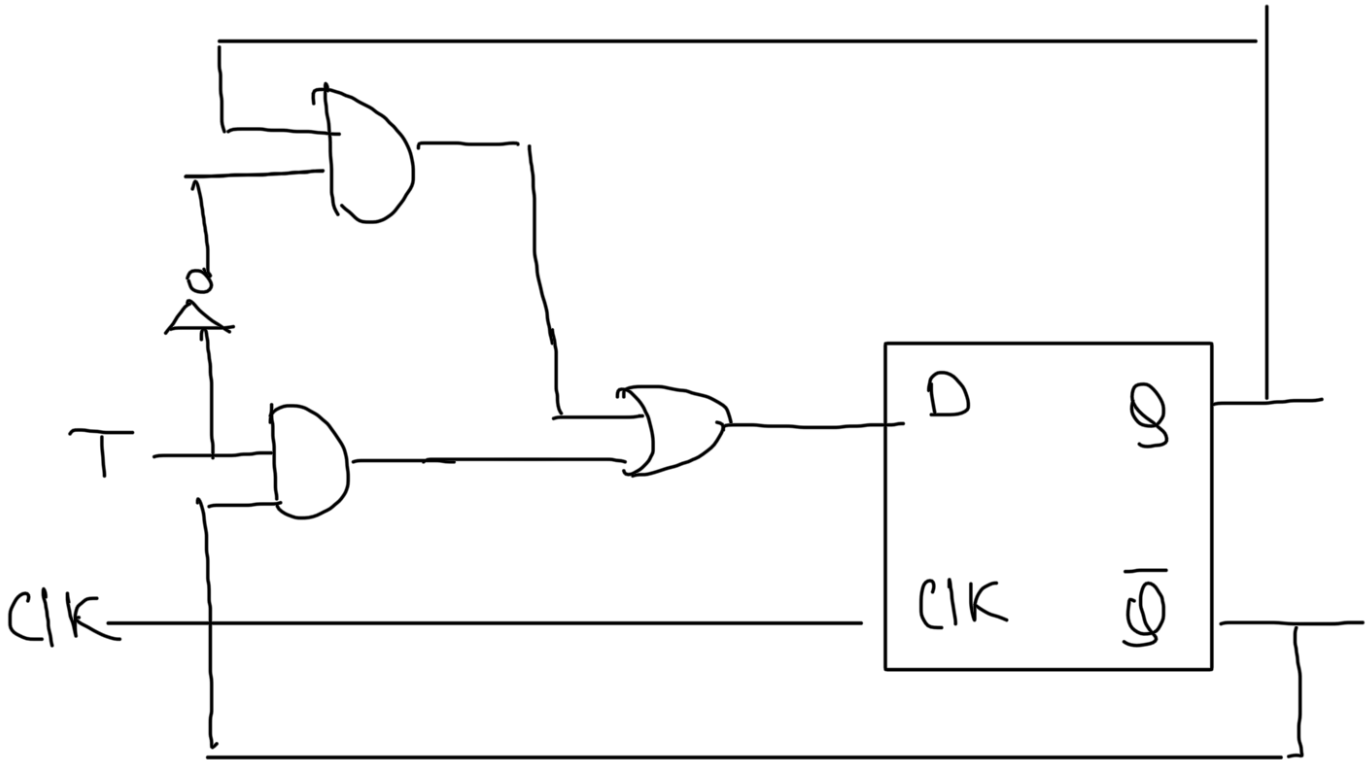
preset = 1 ⇒ no effect

clear = 0 ⇒ Q = 0

clear = 1 ⇒ no effect



T flip flop



T → toggle

T = 0 D = Q → same state

T = 1 → $D = \overline{Q}$

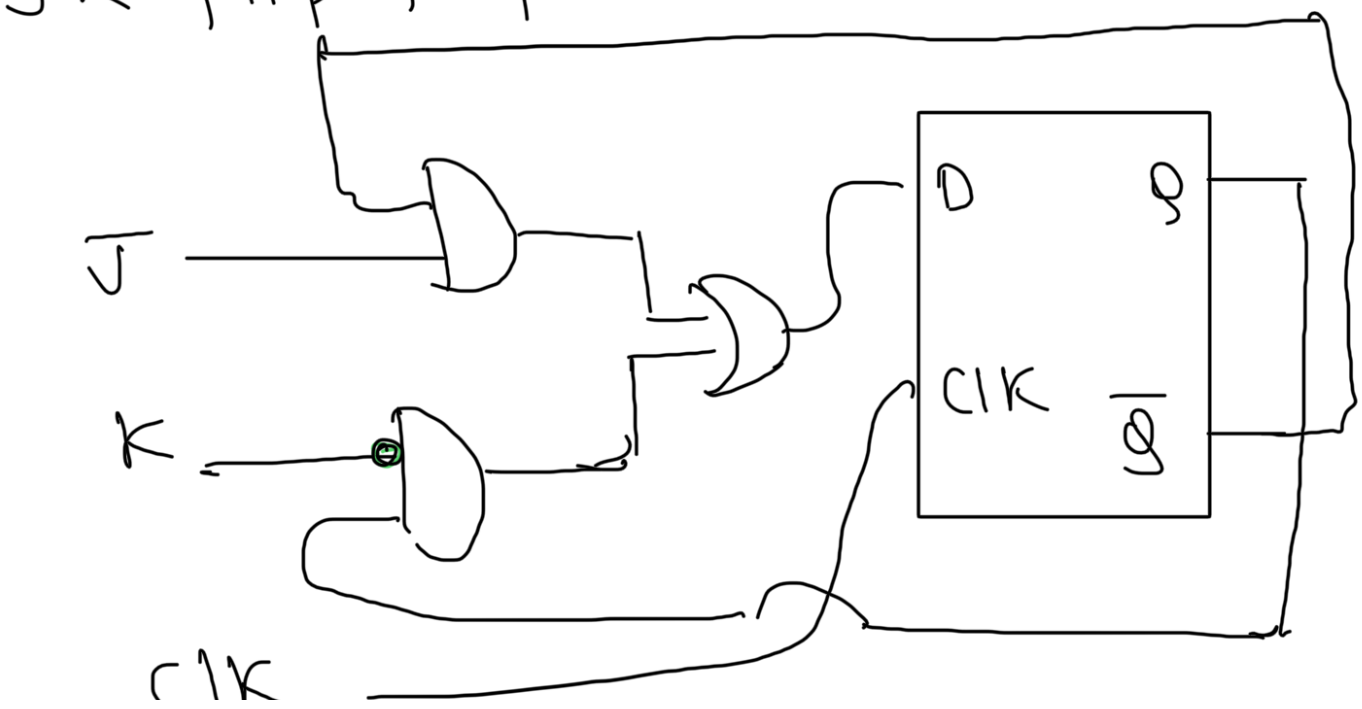
characteristic eqn:

$Q(t+1) = D = TQ' + T'Q$

T	Q(t+1)
0	Q(t)
1	Q'(t)

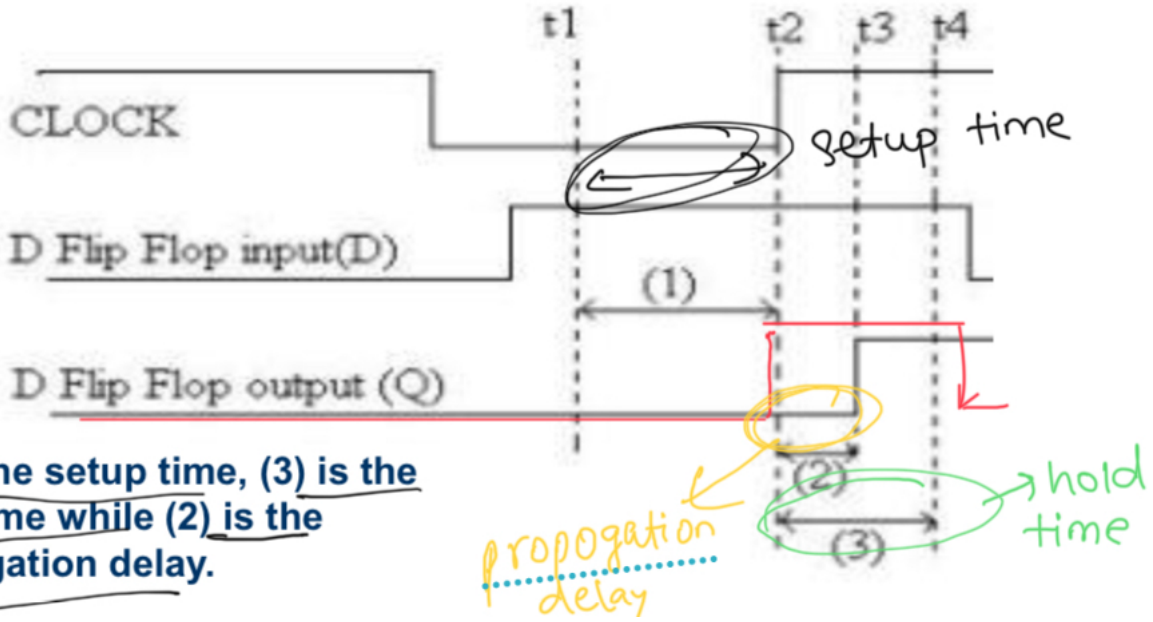
(+ve edge)

JK flip flop



$$D = Q(t+1) = \bar{Q}J + Q\bar{K}$$

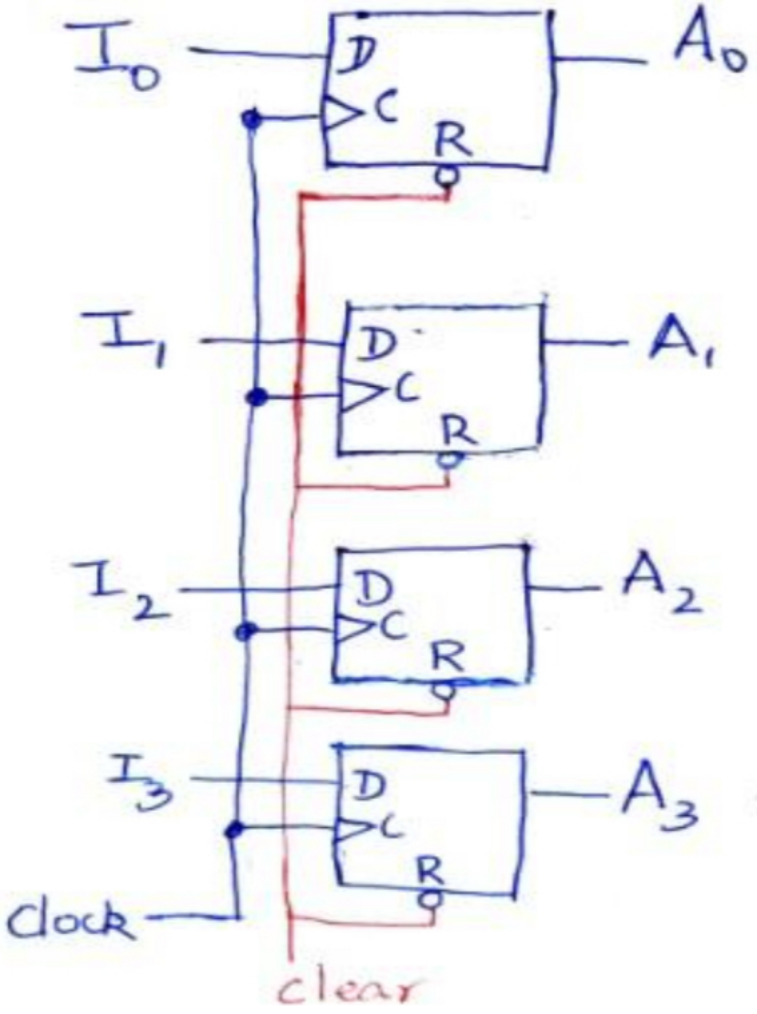
J	K	Q(t+1)
0	0	Q(t)
0	1	0
1	0	1
1	1	Q'(t)



(1) is the setup time, (3) is the hold time while (2) is the propagation delay.

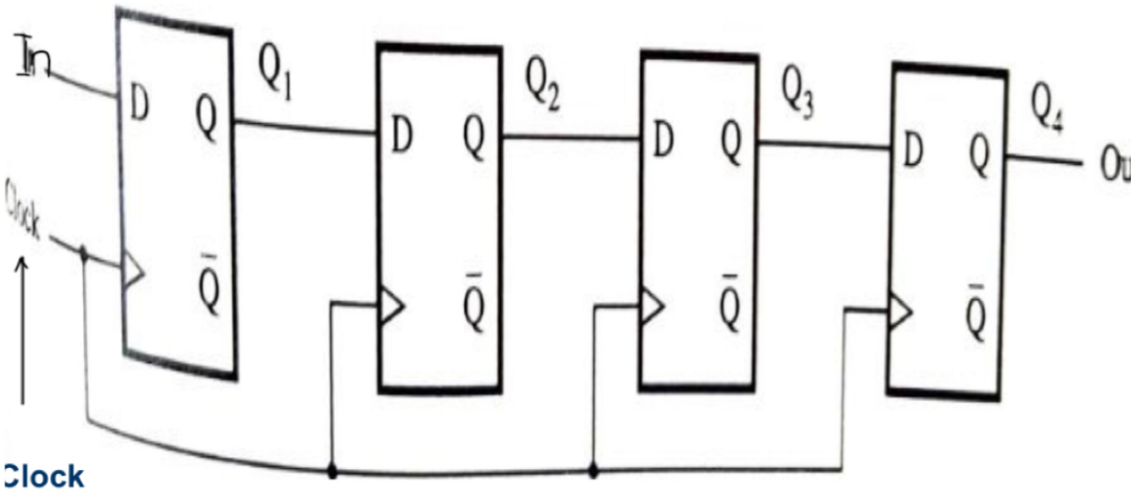
Register: Multiple flip flops

Register with parallel load



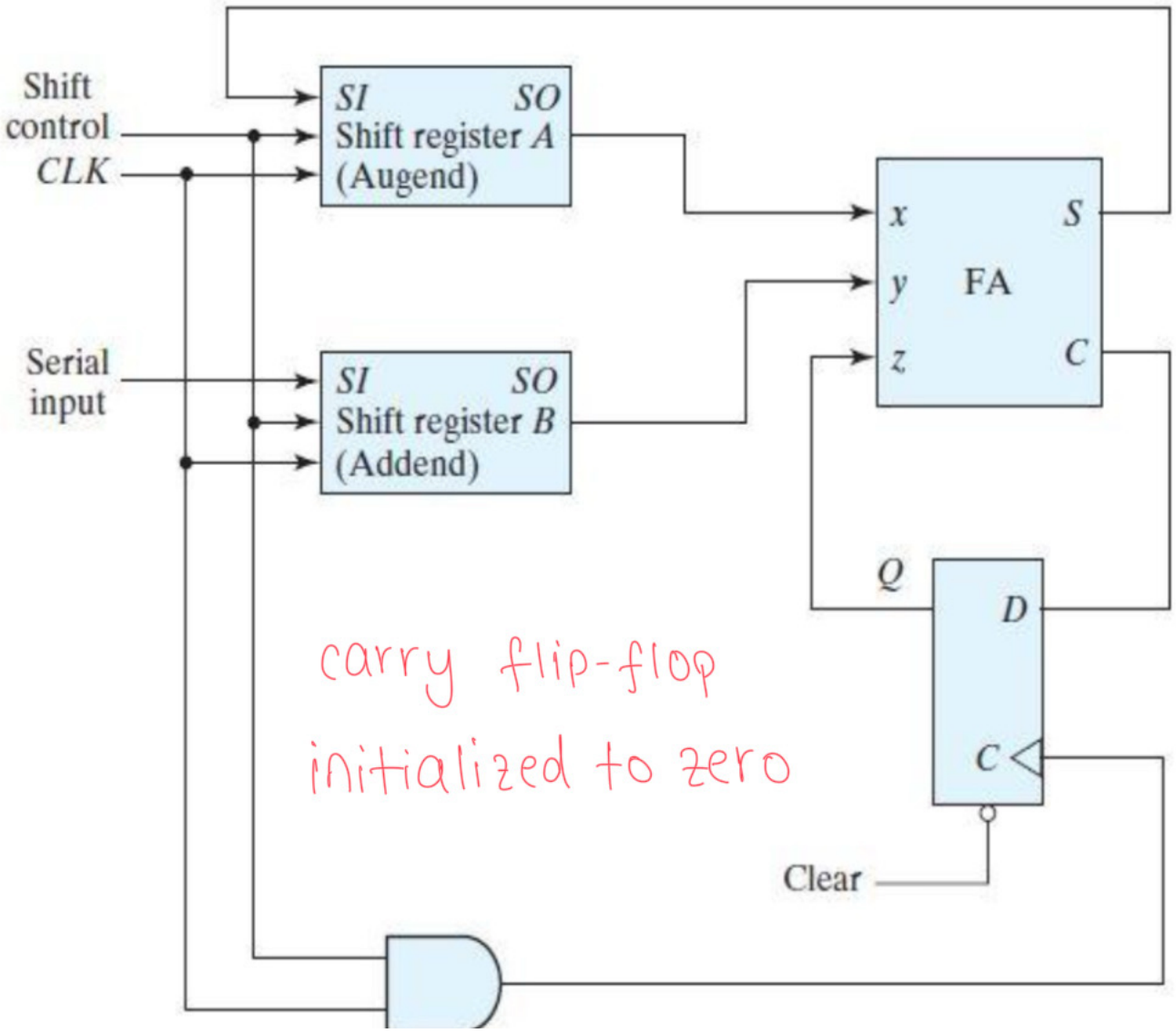
Shift Register

D - flip flop



we need flip flops cuz change in value of In would pass through more than one latch during the time clock = 1

Serial addition:


Shift control
CLK
SI
SO
Shift register A
(Augend)
Serial input
SI
SO
Shift register B
(Addend)
x
y
z
FA
S
C
Q
D
C
Clear
carry flip-flop
initialized to zero

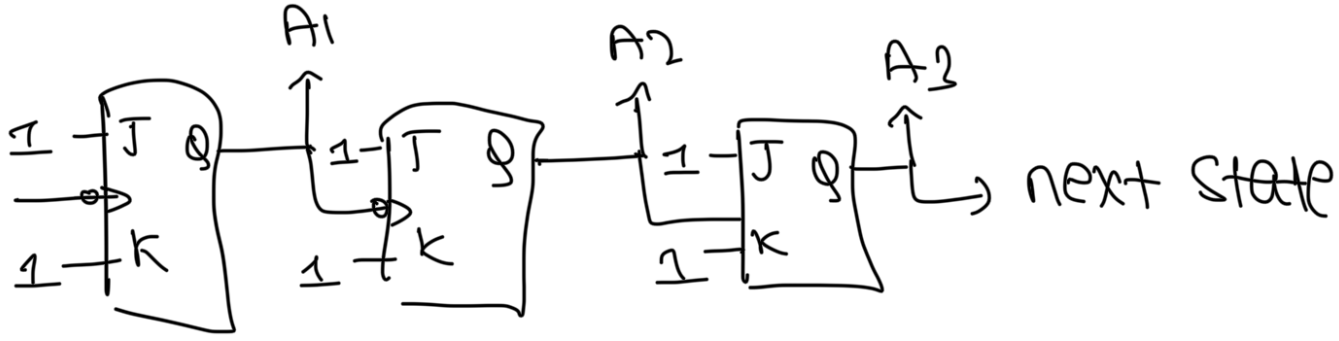
Sum is stored in Augend register itself.

→ Shift control: enables registers & carry flip-flop.

→ we stop addition by disabling control input

→ less hardware than others, but also lesser speed

Ripple counter:

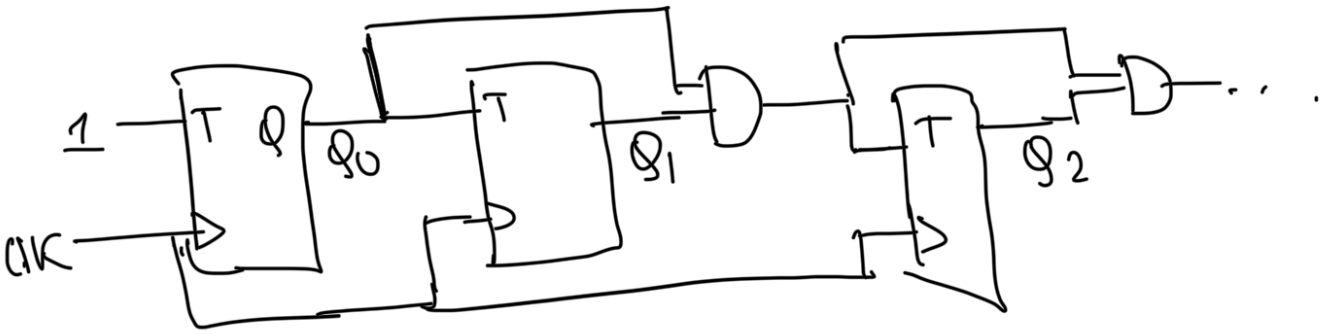


A1 → complemented every time

A2 → complemented when A1 goes from 1→0

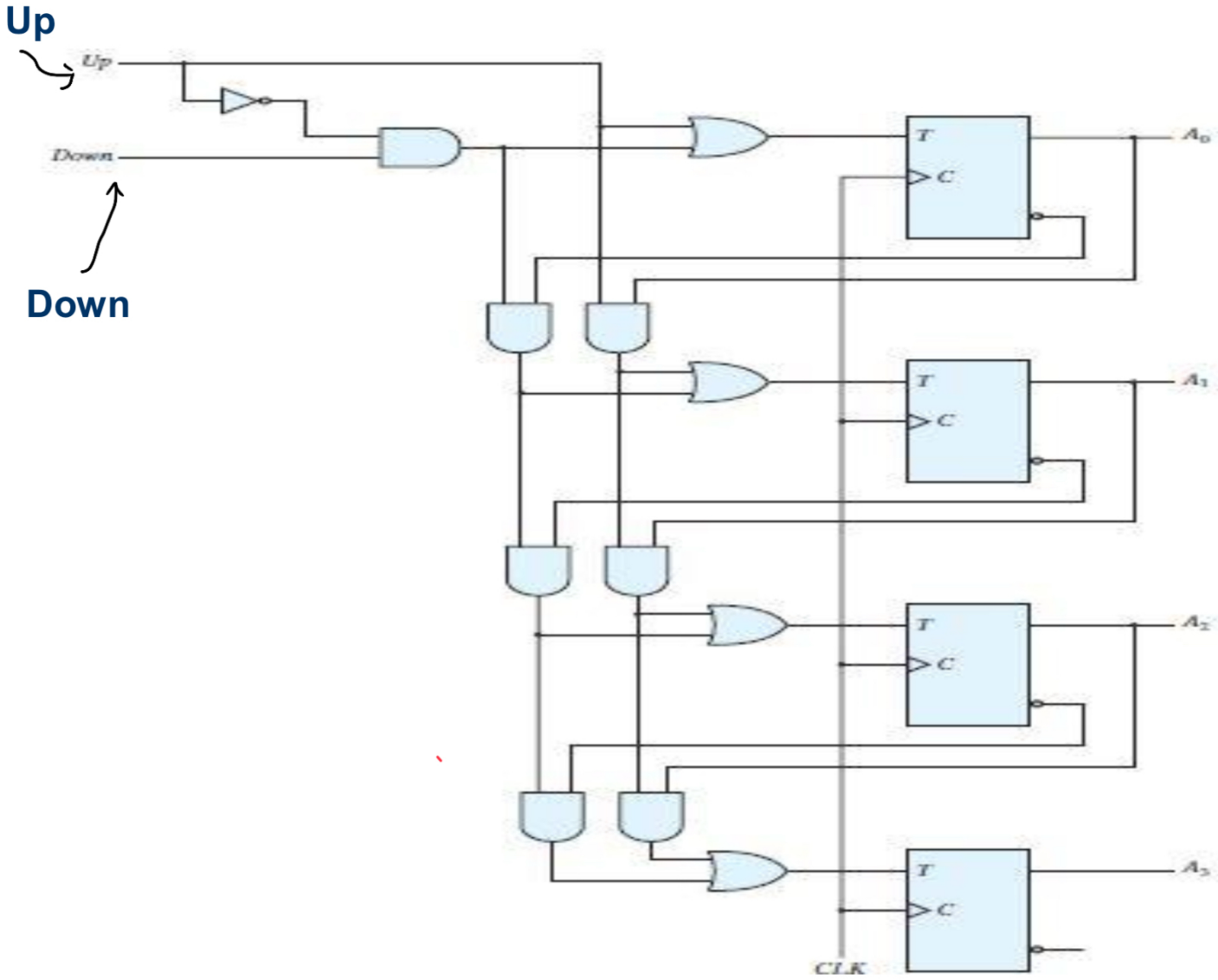
A's → complemented when A_2 goes from 1 → 0

Synchronous up counter:



$T_1 = Q_0$ $T_2 = Q_0Q_1$ $T_3 = Q_0Q_1Q_2$

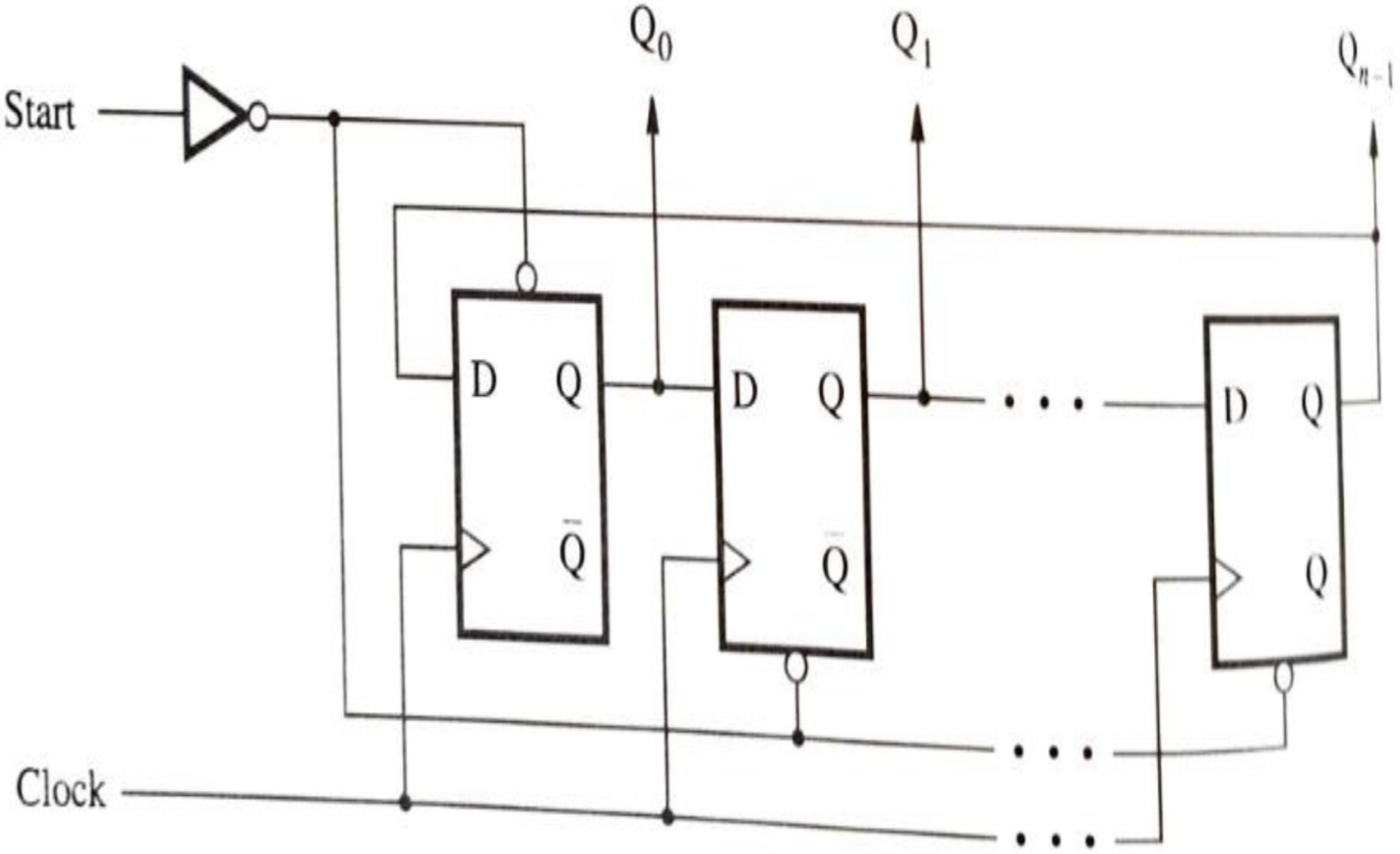
Up-down counter:



UP = 1 ⇒ up counting

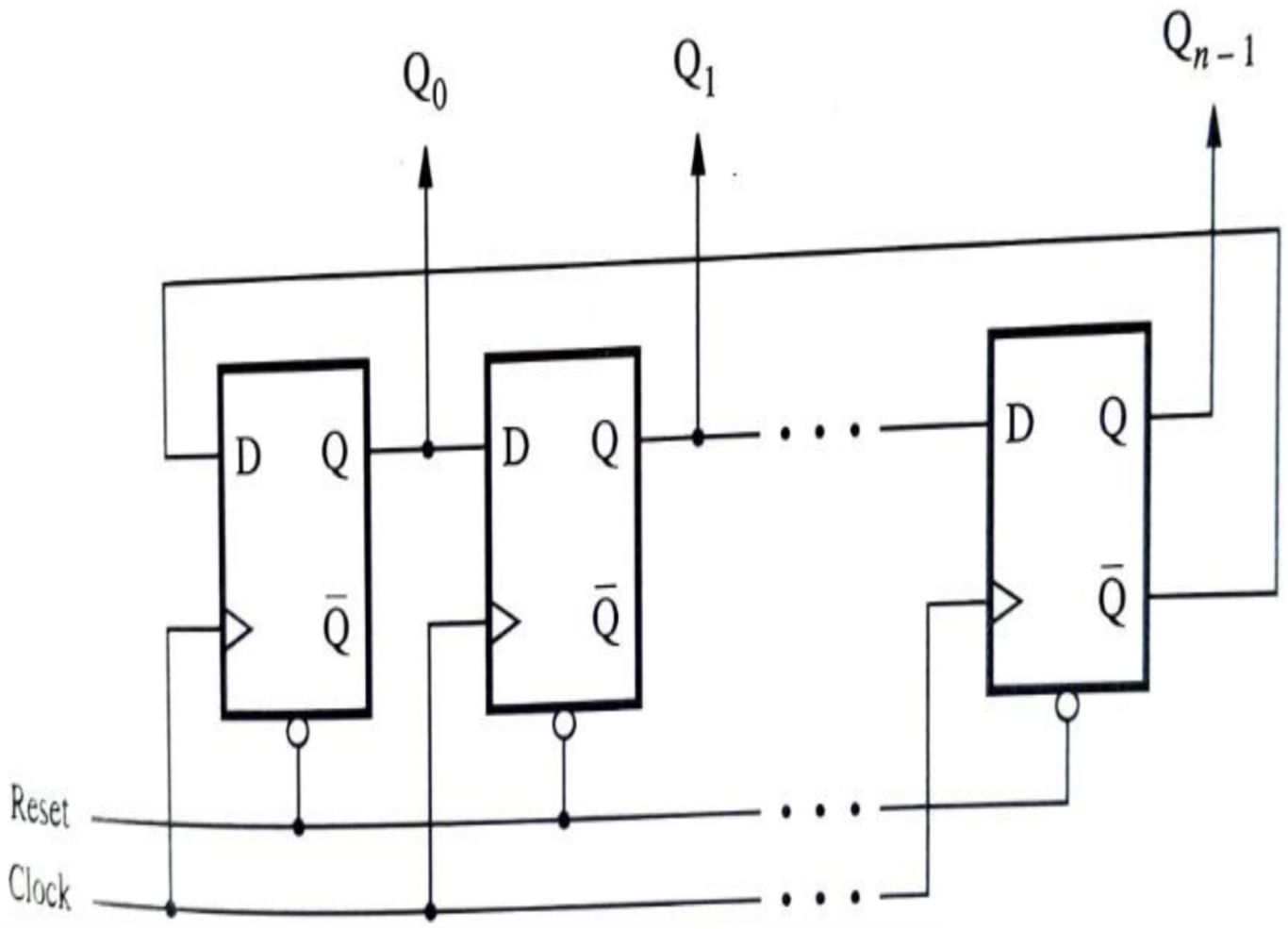
up = 0 , down = 1 ⇒ down counting

Ring counter:



Start presets left most F.F to 1, others to 0 by clr. 1000 → 0100 → 0010 → 0001

Johnson counter

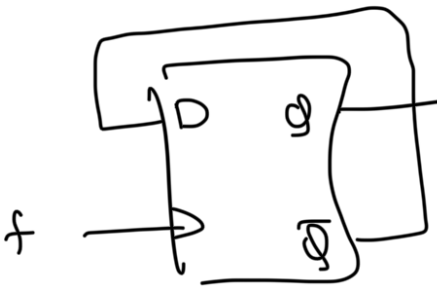

Q_0
Q_1
Q_{n-1}
D
Q
$\bar{Q}$
D
Q
$\bar{Q}$
D
Q
$\bar{Q}$
Reset
Clock

all 0 initially (reset)

0000 → 1000 → 1100 → 1110 → 1111 → 0111 →

0011 → 0001 →

Flip-flops as frequency divider



→ output frequency = f/2
